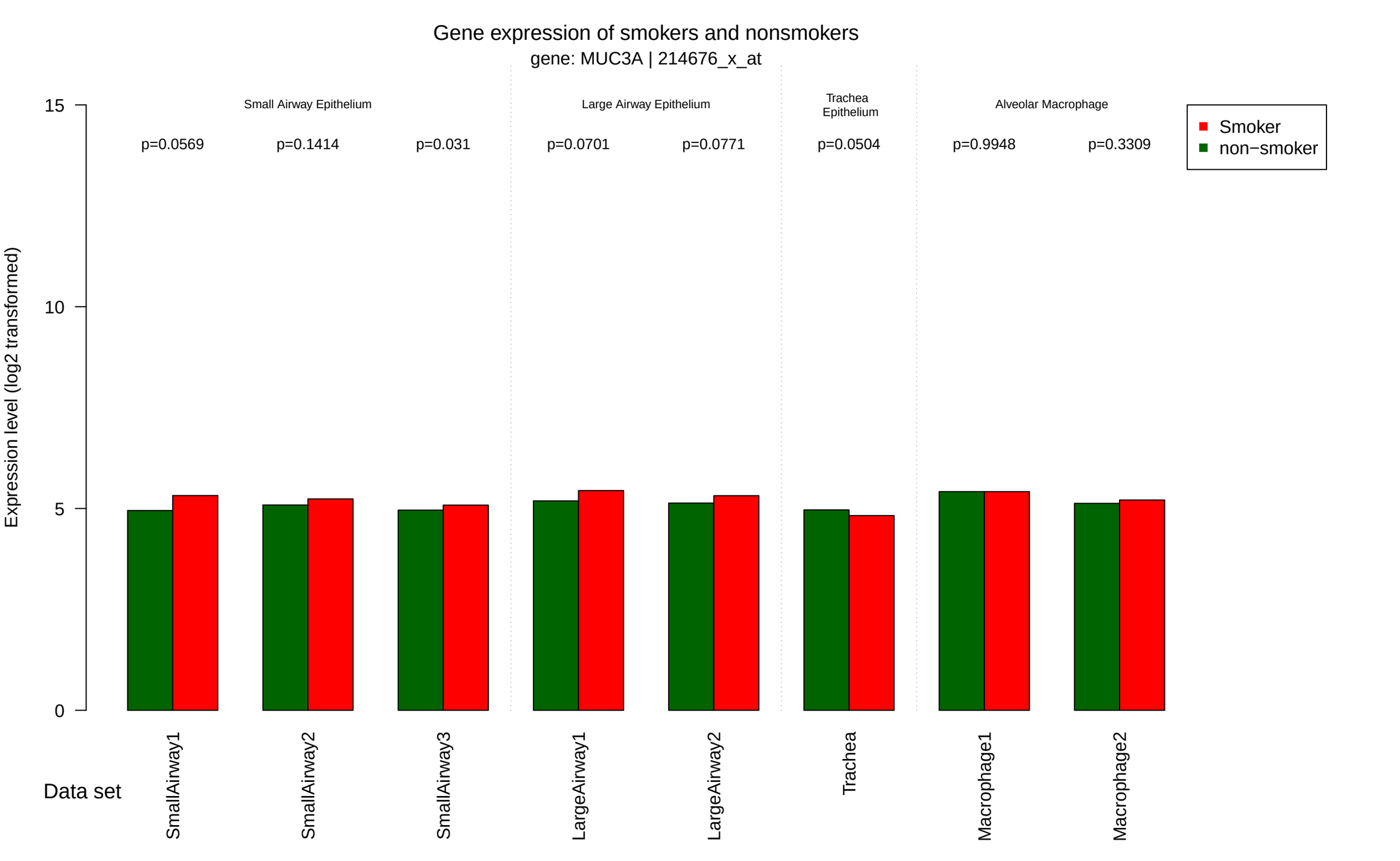
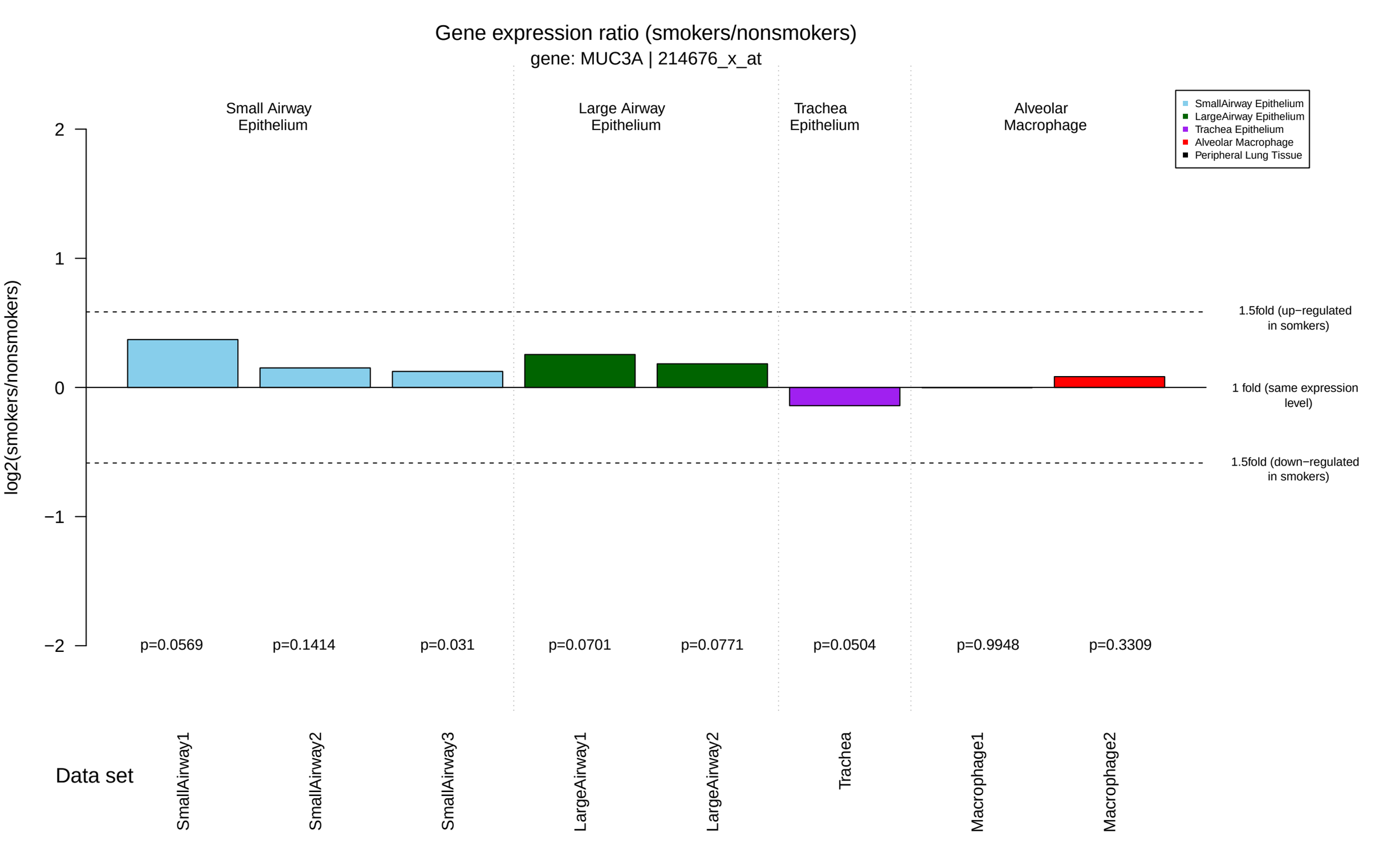
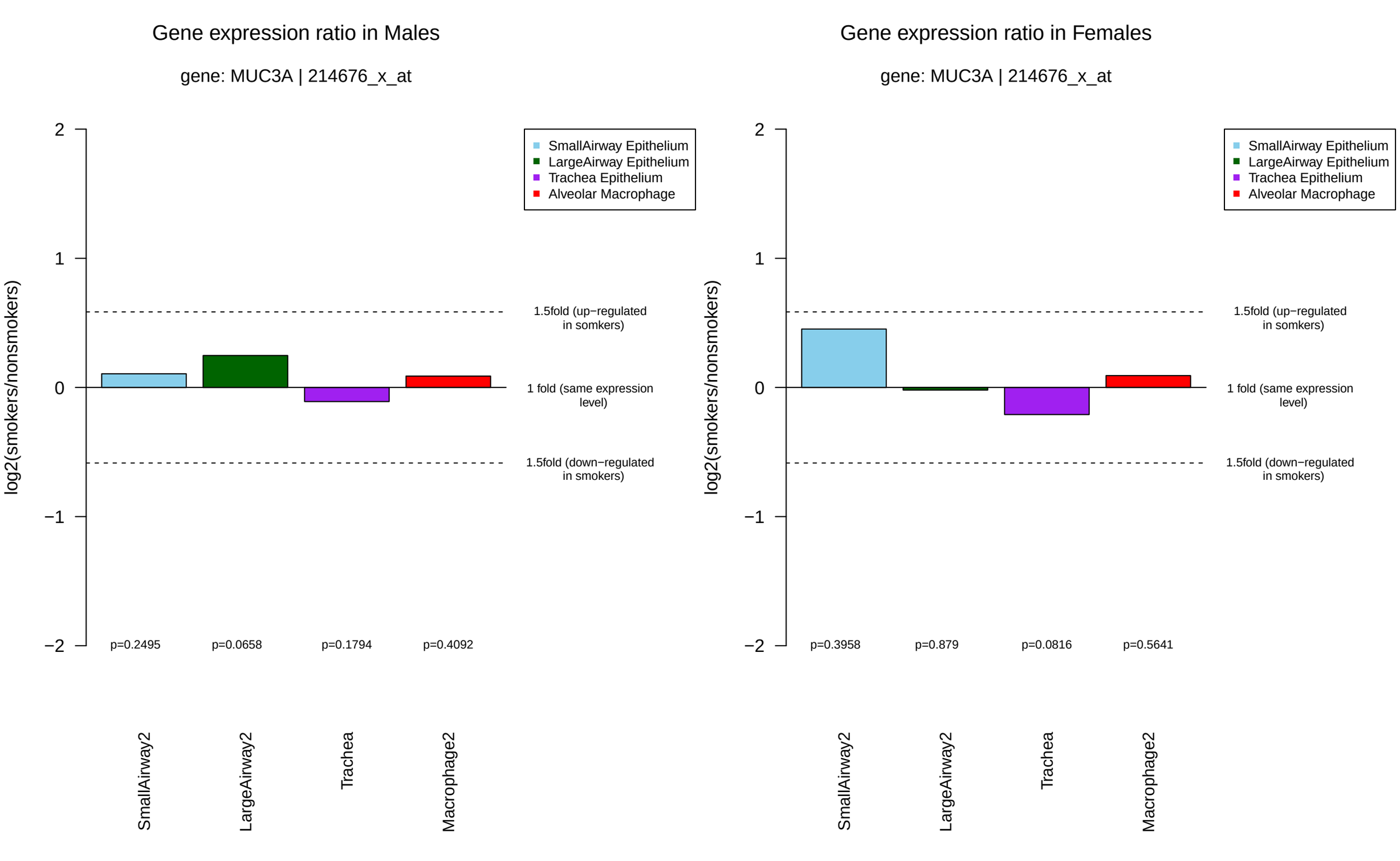
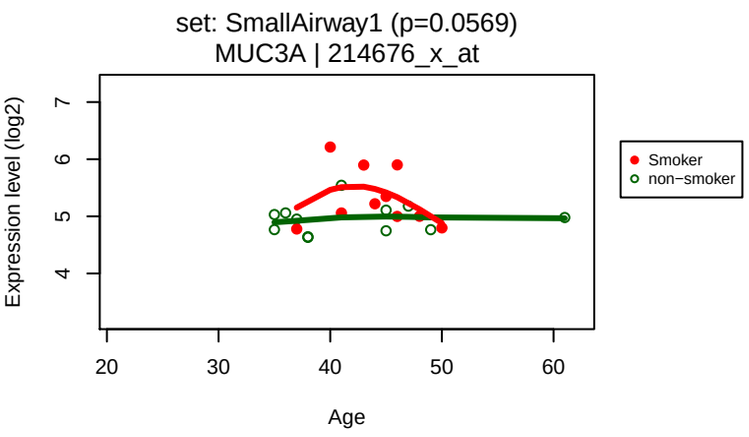
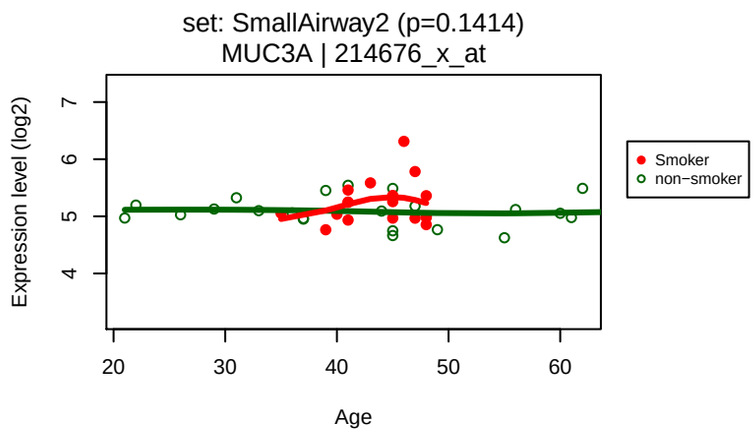
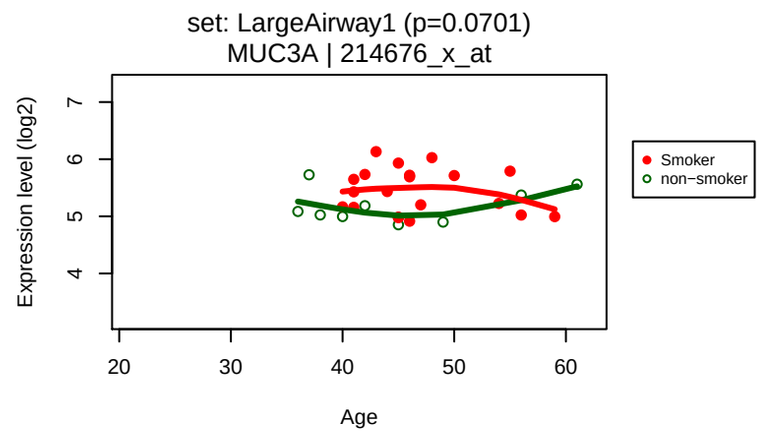

Gene expression of smokers and nonsmokers
gene: MUC3A | 214676_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0569
p=0.1414
p=0.031
p=0.0701
p=0.0771
p=0.0504
p=0.9948
p=0.3309
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

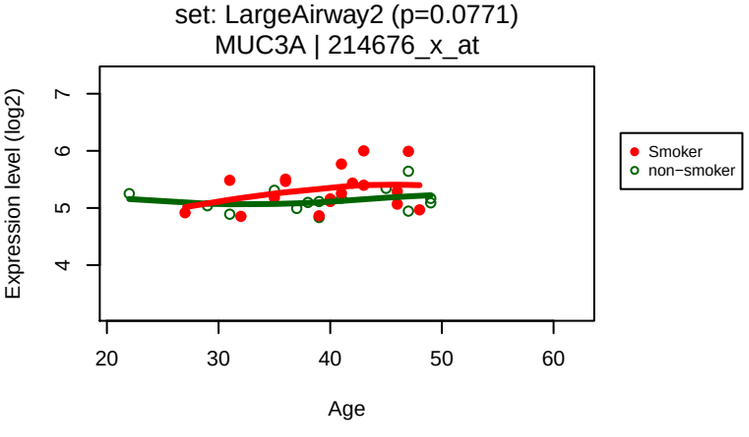

Gene expression ratio (smokers/nonsmokers)
gene: MUC3A | 214676_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0569
p=0.1414
p=0.031
p=0.0701
p=0.0771
p=0.0504
p=0.9948
p=0.3309
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

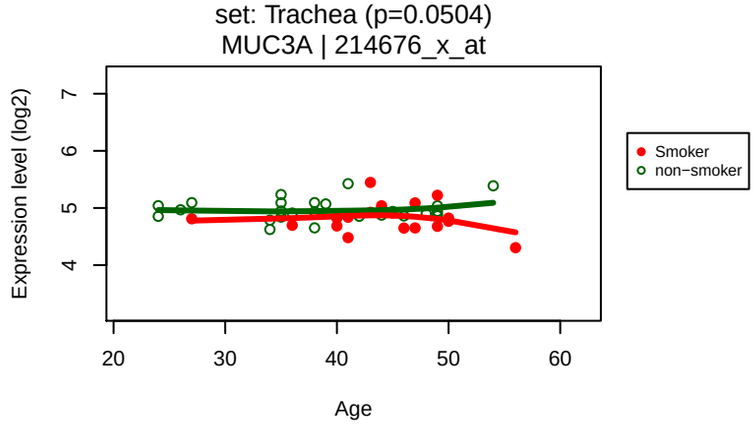

Gene expression ratio in Males
gene: MUC3A | 214676_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2495
p=0.0658
p=0.1794
p=0.4092
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC3A | 214676_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3958
p=0.879
p=0.0816
p=0.5641
SmallAirway2
LargeAirway2
Trachea
Macrophage2

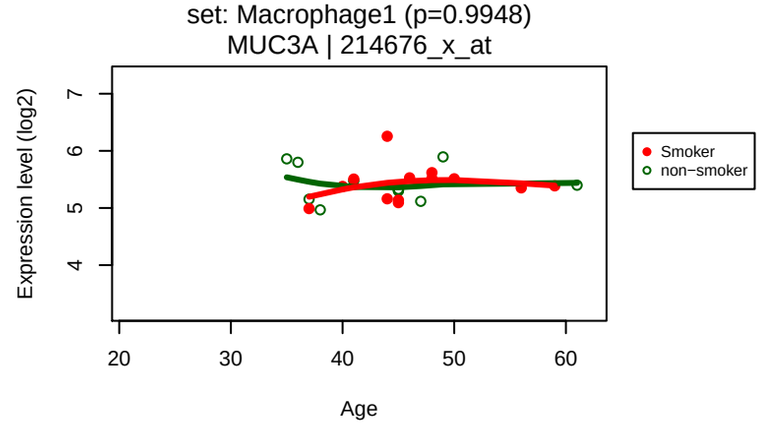

set: SmallAirway1 (p=0.0569)
MUC3A | 214676_x_at
Expression level (log2)
Age
Smoker
non−smoker

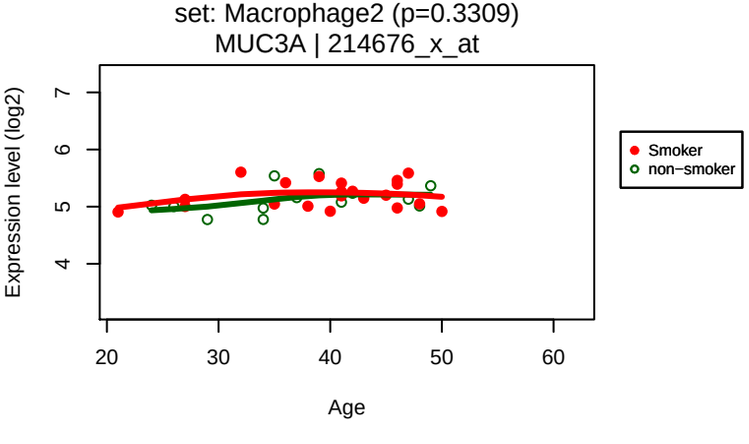

set: SmallAirway2 (p=0.1414)
MUC3A | 214676_x_at
Expression level (log2)
Age
Smoker
non−smoker

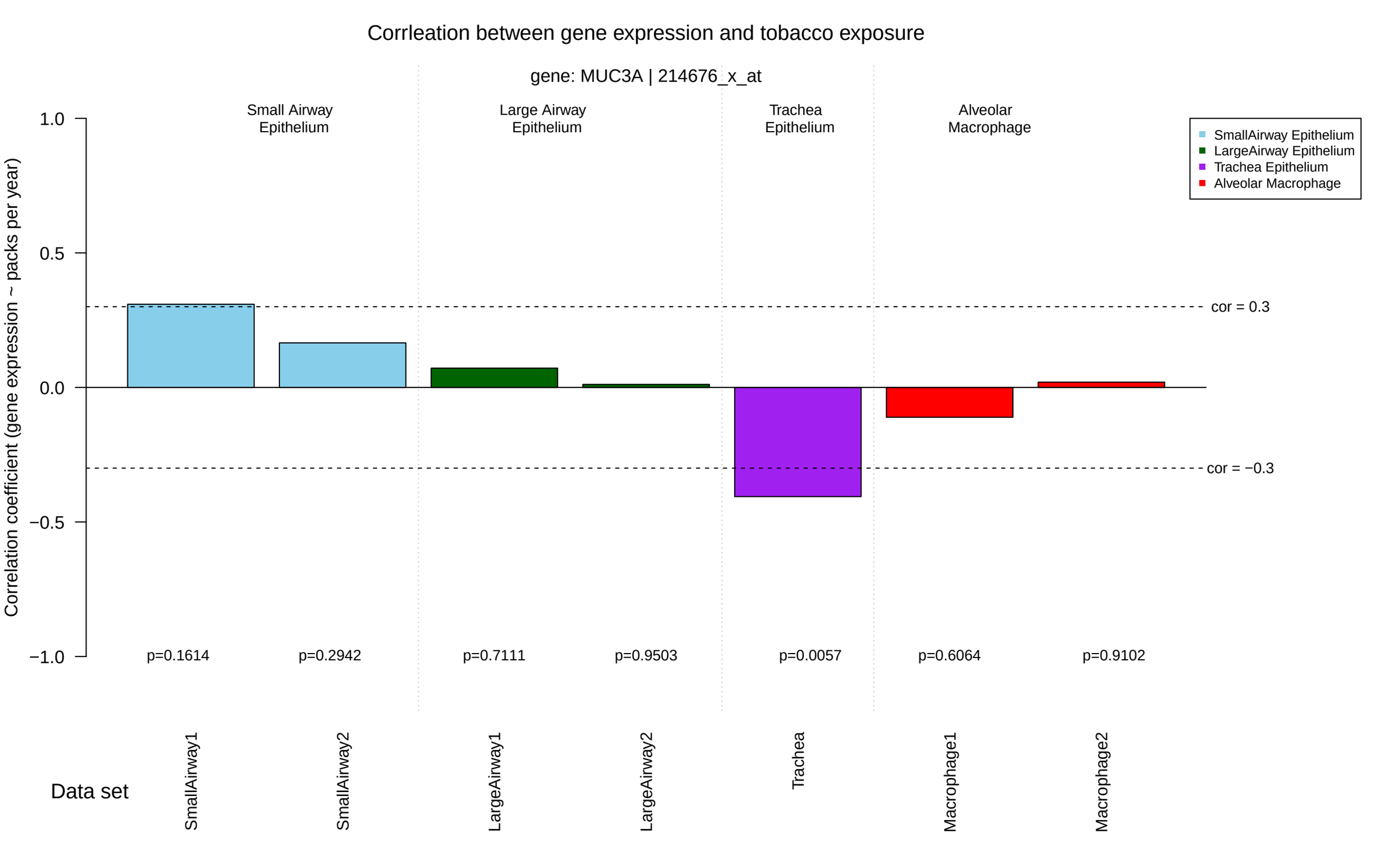

set: LargeAirway1 (p=0.0701)
MUC3A | 214676_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0771)
MUC3A | 214676_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0504)
MUC3A | 214676_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9948)
MUC3A | 214676_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3309)
MUC3A | 214676_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC3A | 214676_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1614
p=0.2942
p=0.7111
p=0.9503
p=0.0057
p=0.6064
p=0.9102
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2